CHANDIGARH UNIVERSITY

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AU4-UNIVERSITY INSTITUTE OF ENGINEERING

COMPUTER SCIENCE ENGINEERING

22SMT-121 MATHEMATICS-1

22BCS401, 22BCS402 & 22BCS416

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UNIT - I

MATRICES

Syllabus for UNIT-I : MATRICES

Matrices, Types of Matrices: Orthogonal Matrices, Rank of matrices, Elementary transformation, reduction to normal form, Consistency and solution of homogenous and non-homogeneous algebraic equations, Eigen values & Eigen Vectors, Linear dependence and independence of vectors, Cayley Hamilton Theorem (without proof).

1. Matrices

A matrix is a **Rectangular Array** (or) **Arrangement of Entries** (or) **Elements displayed in Rows and Columns** put within a **Square Bracket [ ]**

In general, the entries of a matrix may be **Real** or **Complex Numbers** or **Functions of one variable (such as Polynomials, Trigonometric functions or a combination of them)** or **more variables** or **any other object**.

Usually, matrices are denoted by **CAPITAL LETTERS** A, B, C, … etc

General form of a Matrix

If a matrix A has m rows and n columns, then it is written as

$$A=[a_{ij}]_{m\times n}\ ,1\leq i\leq m,1\leq j\leq n.$$

General form of a Matrix

$$
A=[a_{ij}]_{m\times n}=
\begin{array}{ccc}
\multicolumn{2}{c}{\text{n columns}} & \\
\begin{array}{cccccc}\text{Column 1} & \text{Column 2} & & \text{Column j} & & \text{Column n}\\ \downarrow & \downarrow & & \downarrow & & \downarrow\end{array} & & \\
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n}\\
a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n}\\
\vdots & \vdots & \ddots & \vdots & \vdots & \vdots\\
a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in}\\
\vdots & \vdots & \vdots & \vdots & \ddots & \vdots\\
a_{m1} & a_{m2} & \cdots & a_{mj} & \cdots & a_{mn}
\end{bmatrix}
&
\begin{array}{l}
\leftarrow \text{Row 1}\\
\leftarrow \text{Row 2}\\
\\
\leftarrow \text{Row i}\\
\\
\leftarrow \text{Row m}
\end{array}
& \text{m rows}
\end{array}
$$

Note: m and n are positive integers

Examples of a Matrix

$$A=\begin{bmatrix}2 & 0 & -1\\ 1 & 4 & 5\\ 9 & -8 & 6\end{bmatrix},\quad B=\begin{bmatrix}7 & -9 & 1.2 & 0\\ \sin\frac{x}{4} & 2 & x^2 & 4\\ \cos\frac{x}{2} & 1 & 3 & -6\end{bmatrix},\text{ and }\quad C=\begin{bmatrix}1 & 5 & -7\\ 3.4 & \frac{1}{2} & \sqrt{2}\\ e^2 & -3 & 4\\ \sqrt{5} & 2 & a\end{bmatrix}.$$

In a matrix,
•*HORIZONTAL LINES* of elements are known as **ROWS**
•*VERTICAL LINES* of elements are known as **COLUMNS**.

Thus
A has 3 rows and 3 columns,
B has 3 rows and 4 columns, and
C has 4 rows and 3 columns

Order of a Matrix

If a matrix A has m rows and n columns then the **order or size** of the matrix A is defined to be $m \times n$ (read as m by n).

The objects $a_{11}, a_{12}, \ldots, a_{mn}$ are called **elements or entries** of the matrix $A = [a_{ij}]_{m \times n}$. The element a_{ij} is common to i^{th} row and j^{th} column and is called $(i, j)^{th}$ element of A. Observe that the i^{th} row and j^{th} column of A are $1 \times n$ and $m \times 1$ matrices respectively and are given by $[a_{i1} \ a_{i2} \ \ldots \ a_{in}]$ and $\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$

1. APPLICATIONS OF MATRICES

Generally, a **MATRIX** is nothing but a **Rectangular Array of objects**.

These matrices can be **Visualised** in **Day-To-Day Applications** where we use matrices to represent a **Military Parade**, **School Assembly**, **Vegetation**, **Class room** **etc**.

Vegetation

Military Parade

School Assembly

Classroom

Engineering & Technology

In Cryptography: In encryption, we use it to very sensitive data for security purposes to **Encode And To Decode** this data we need matrices. There is a key that helps encode and decode data which is generated by matrices.

In Animation: It can help make animations more **PRECISE AND PERFECT**.

In Games Especially 3D: They use it to **ALTER THE OBJECT**, in 3D space. They use the **3D Matrix to 2D Matrix** to convert it into the different objects as per requirement.

Engineering & Technology

In IT companies: Use **Matrices** as Data Structures to **Track User Information**, perform **Search Queries**, and **Manage Databases**. In the world of information security, many systems are designed to work with matrices.

- Matrices are very **important and absolutely necessary** in handling system of linear equations which arise as mathematical models of real-world problems.

- Mathematicians **Gauss, Jordan, Cayley, and Hamilton** have developed the **THEORY OF MATRICES** which has been used in investigating solutions of **Systems of Linear Equations**.

Example:

Consider the marks scored by a student in different subjects and in different terminal examinations. They are exhibited in a tabular form as given below :

	Tamil	English	Mathematics	Science	Social Science
Exam 1	48	71	80	62	55
Exam 2	70	68	91	73	60
Exam 3	77	84	95	82	62

This tabulation represents the above information in the form of matrix. What does the entry in the third row and second column represent?

The above information may be represented in the form of a 3×5 matrix A as

$$A = \begin{bmatrix} 48 & 71 & 80 & 62 & 55 \\ 70 & 68 & 91 & 73 & 60 \\ 77 & 84 & 95 & 82 & 62 \end{bmatrix}.$$

The entry 84 common to the third row and the second column in the matrix represents the mark scored by the student in English Exam 3.

TYPES OF MATRICES

Row Matrix

A matrix having only one row is called a **row matrix.**

For instance, $A = [A]_{1\times 4} = [1 \;\; 0 \;\; -1.1 \;\; \sqrt{2}]$ is a row matrix. More generally, $A = [a_{ij}]_{1\times n} = [a_{1j}]_{1\times n}$ is a row matrix of order $1 \times n$.

Column Matrix

A matrix having only one column is called a **column matrix.**

For instance, $[A]_{4\times 1} = \begin{bmatrix} x+1 \\ x^2 \\ 3x \\ 4 \end{bmatrix}$ is a column matrix whose entries are real valued functions of real variable x. More generally, $A = [a_{ij}]_{m\times 1} = [a_{i1}]_{m\times 1}$ is a column matrix of order $m \times 1$.

Zero Matrix

A matrix $A=[a_{ij}]_{m\times n}$ is said to be a **zero matrix** or **null matrix** or **void matrix** denoted by O if $a_{ij}=0$ for all values of $1\le i\le m$ and $1\le j\le n$.

For instance, $[0]$, $\begin{bmatrix}0&0&0\\0&0&0\\0&0&0\end{bmatrix}$ and $\begin{bmatrix}0&0&0&0\\0&0&0&0\end{bmatrix}$ are zero matrices of order 1×1, 3×3 and 2×4 respectively.

A matrix A is said to be a **non-zero matrix** if at least one of the entries of A is non-zero.

Square Matrix

A matrix in which number of rows is equal to the number of columns, is called a **square matrix**. That is, a matrix of order $n \times n$ is often referred to as a square matrix of order n.

For instance, $A = \begin{bmatrix} a & b & c \\ d & c & f \\ g & h & l \end{bmatrix}$ is a square matrix of order 3.

Principle diagonal

In a square matrix $A = [a_{ij}]_{n\times n}$ of order n, the elements $a_{11}, a_{22}, a_{33}, \ldots, a_{nn}$ are called the **principal diagonal** or simply the **diagonal** or **main diagonal** or **leading diagonal elements.**

Diagonal Matrix

A square matrix $A = [a_{ij}]_{n \times n}$ is called a **diagonal matrix** if $a_{ij} = 0$ whenever $i \neq j$.

Thus, in a diagonal matrix all the entries except the entries along the main diagonal are zero. For instance,

$$A = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & 0.5 \end{bmatrix},\ B = \begin{bmatrix} r & 0 \\ 0 & s \end{bmatrix},\ C = [6],\ \text{and}\ D = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & a_{nn} \end{bmatrix}$$

are diagonal matrices of order 3, 2, 1, and n respectively.

Scalar Matrix

A diagonal matrix whose entries along the principal diagonal are equal, is called a Scalar matrix.

That is, a square matrix $A = [a_{ij}]_{n\times n}$ is said to be a scalar matrix if $a_{ij} = \begin{cases} c & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

where c is a fixed number. For instance,

$$A = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix}, B = \begin{bmatrix} -5 & 0 \\ 0 & -5 \end{bmatrix}, C = \begin{bmatrix} \sqrt{3} \end{bmatrix} \text{ and } D = \begin{array}{c} \begin{bmatrix} c & 0 & \cdots & 0 \\ 0 & c & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c \end{bmatrix} \end{array} \begin{array}{l} R_1 \\ R_2 \\ \\ R_n \end{array}$$

are scalar matrices of order 3, 2, 1, and n respectively.

Observe that any square zero matrix can be considered as a scalar matrix with scalar 0.

Unit Matrix

A square matrix in which all the diagonal entries are 1 and the rest are all zero is called a **unit matrix**. Thus, a square matrix $A = [a_{ij}]_{n\times n}$ is said to be a unit matrix if $a_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$.

We represent the unit matrix of order n as I_n. For instance,

$$I_1 = [1], I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ and } I_n = \begin{array}{c} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \end{array} \begin{array}{l} R_1 \\ R_2 \\ \\ R_n \end{array}$$

are unit matrices of order 1, 2, 3 and n respectively.

There are two kinds of triangular matrices namely upper triangular and lower triangular matrices.

Upper Triangular Matrix

A square matrix is said to be an upper triangular matrix if all the elements below the main diagonal are zero.

Thus, the square matrix $A = [a_{ij}]_{n\times n}$ is said to be an upper triangular matrix if $a_{ij} = 0$ for all $i > j$. For instance,

$$\begin{bmatrix} 4 & 3 & 0 \\ 0 & 7 & 8 \\ 0 & 0 & 2 \end{bmatrix}, \begin{bmatrix} -5 & 2 \\ 0 & 1 \end{bmatrix}, \text{ and } \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & a_{nn} \end{bmatrix}$$ are all upper triangular matrices.

Lower Triangular Matrix

A square matrix is said to be a lower triangular matrix if all the elements above the main diagonal are zero.

More precisely, a square matrix $A = [a_{ij}]_{n \times n}$ is said to be a lower triangular matrix if $a_{ij} = 0$ for all $i < j$. For instance,

$$\begin{bmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 4 & 1 & 0 \\ 8 & -5 & 7 \end{bmatrix}, \begin{bmatrix} -2 & 0 \\ 9 & -3 \end{bmatrix}, \text{ and } \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$ are all lower triangular matrices.

Triangular Matrix

A square matrix which is either upper triangular or lower triangular is called a **triangular matrix.**

Observe that a square matrix that is both upper and lower triangular simultaneously will turn out to be a diagonal matrix.

$$A = \begin{bmatrix} 2.5 & 0 & 0 \\ 0 & \sqrt{3} & 0 \\ 0 & 0 & 0.5 \end{bmatrix}, \quad B = \begin{bmatrix} r & 0 \\ 0 & s \end{bmatrix}$$

Problems based on Matrix Addition, Subtraction and Multiplication

Problem-1

Compute $A+B$ and $A-B$ if

$$A=\begin{bmatrix} 4 & \sqrt{5} & 7 \\ -1 & 0 & 0.5 \end{bmatrix} \text{ and } B=\begin{bmatrix} \sqrt{3} & \sqrt{5} & 7.3 \\ 1 & \frac{1}{3} & \frac{1}{4} \end{bmatrix}.$$

Solution

By the definitions of addition and subtraction of matrices, we have

$$A+B=\begin{bmatrix} 4+\sqrt{3} & 2\sqrt{5} & 14.3 \\ 0 & \frac{1}{3} & \frac{3}{4} \end{bmatrix} \text{ and } A-B=\begin{bmatrix} 4-\sqrt{3} & 0 & -0.3 \\ -2 & -\frac{1}{3} & \frac{1}{4} \end{bmatrix}$$

Problem-2

Determine $3B + 4C - D$ if B, C, and D are given by

$$B = \begin{bmatrix} 2 & 3 & 0 \\ 1 & -1 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & -2 & 3 \\ -1 & 0 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 4 & -1 \\ 5 & 6 & -5 \end{bmatrix}.$$

Solution

$$3B + 4C - D = \begin{bmatrix} 6 & 9 & 0 \\ 3 & -3 & 15 \end{bmatrix} + \begin{bmatrix} -4 & -8 & 12 \\ -4 & 0 & 8 \end{bmatrix} + \begin{bmatrix} 0 & -4 & 1 \\ -5 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 13 \\ -6 & -9 & 28 \end{bmatrix}.$$

Problem-3

Solve for x if $\begin{bmatrix} x & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ -1 & -4 & 1 \\ -1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix} = O.$

Solution

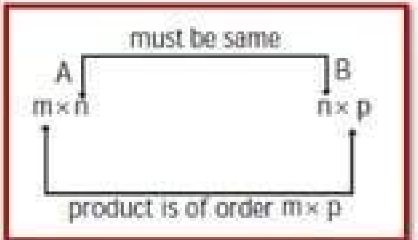
$$\begin{bmatrix} x & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ -1 & -4 & 1 \\ -1 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix} = O$$

That is, $$\begin{bmatrix} x-2+1 & x-8+1 & 2x+2+2 \end{bmatrix} \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix} = O$$

$$\begin{bmatrix} x-1 & x-7 & 2x+4 \end{bmatrix} \begin{bmatrix} x \\ 2 \\ 1 \end{bmatrix} = O$$

$$x(x-1)+2(x-7)+1(2x+4) = 0$$

$$x^2+3x-10 = 0 \Rightarrow x = -5, 2.$$



2. CAYLEY HAMILTON THEOREM

Characteristic Equations

If A is a square matrix of order 3, then its characteristic equation can be written as

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{ where}$$

S_1 = sum of the main diagonal elements = $a_{11} + a_{22} + a_{33}$

S_2 = sum of the minors of main diagonal elements

$$= \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

S_3 = Determinant value of A = $|A|$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Problem-1

Find the characteristic equation of the matrix $\begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$

Solution:

The characteristic equation of A is $\lambda^2 - S_1\lambda + S_2 = 0$

S_1 = Sum of the main diagonal elements $= 1 + 2 = 3$

$S_2 = |A| = \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$

Hence, the required characteristic equation is $\lambda^2 - (3)\lambda + (2) = 0$

i.e., $\lambda^2 - 3\lambda + 2 = 0$

Problem-2

Find the characteristic equation of $\begin{bmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{bmatrix}$

Solution:

Given : matrix is a 3×3 matrix.

$\therefore$ The characteristic equation of A is $\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$

S_1 = Sum of the main diagonal elements = $(2) + (1) + (-4) = -1$

S_2 = Sum of the minors of main diagonal elements

$$= \begin{vmatrix} 1 & 3 \\ 2 & -4 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ -5 & -4 \end{vmatrix} + \begin{vmatrix} 2 & -3 \\ 3 & 1 \end{vmatrix}$$

$$= (-4 - 6) + (-8 + 5) + (2 + 9)$$

$$= -10 - 3 + 11 = -2$$

$$S_3 = |A| = \begin{vmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{vmatrix}$$

$$= 2(-4 - 6) - (-3)(-12 + 15) + 1(6 + 5)$$

$$= 2(-10) + 3(3) + 1(11) = -20 + 9 + 11 = 0$$

Hence, the required characteristic equation is

$$\lambda^3 - (-1)\lambda^2 + (-2)\lambda + (0) = 0$$

$$\lambda^3 + \lambda^2 - 2\lambda = 0$$

Home Works

I. Find the characteristic equation of

		Answers			Answers
1.	$\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$	$\lambda^2 - 4\lambda + 5 = 0$	2.	$\begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix}$	$\lambda^2 + \lambda - 6 = 0$
3.	$\begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$	$\lambda^2 - 3\lambda + 2 = 0$	4.	$\begin{bmatrix} 1 & 1 \\ 3 & -1 \end{bmatrix}$	$\lambda^2 - 4 = 0$
5.	$\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$	$\lambda^2 - 5\lambda + 6 = 0$	6.	$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	$\lambda^3 - 18\lambda^2 + 45\lambda = 0$
7.	$\begin{bmatrix} 3 & 2 & -1 \\ 2 & 1 & 0 \\ 4 & -1 & 6 \end{bmatrix}$	$\lambda^3 - 10\lambda^2 + 27\lambda = 0$	8.	$\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	$\lambda^3 - 7\lambda^2 + 36 = 0$

Cayley Hamilton Theorem

Statement:

Every Square matrix satisfies its own characteristic equations

Uses of Cayley Hamilton Theorem

1. To calculate the positive integral power of matrix A.

2. To calculate the inverse of a non-singular matrix A.

Problem-3

Show that the matrix $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ satisfies its own characteristic equation.

Sol. Let $A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

The characteristic equation of the given matrix is $|A - \lambda I| = 0$

i.e., $\lambda^2 - S_1\lambda + S_2 = 0$

where S_1 = sum of the main diagonal elements

$= 1 + 1 = 2$

$S_2 = |A| = \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 1 + 4 = 5$

$\therefore$ The characteristic equation is $\lambda^2 - 2\lambda + 5 = 0$

To prove : $A^2 - 2A + 5I = O$

$$A^2 = AA = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & -4 \\ 4 & -3 \end{pmatrix}$$

$$A^2 - 2A + 5I = \begin{pmatrix} -3 & -4 \\ 4 & -3 \end{pmatrix} - 2\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} + 5\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & -4 \\ 4 & -3 \end{pmatrix} + \begin{pmatrix} -2 & 4 \\ -4 & -2 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O$$

Therefore, the given matrix satisfies its own characteristic equation.

Problem-4

Verify Cayley Hamilton Theorem find A^4 and A^{-1} when

$$A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Sol. The characteristic equation of A is $|A - \lambda I| = 0$

$$\text{i.e., } \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \text{ where}$$

S_1 = sum of its leading diagonal elements $= 2 + 2 + 2 = 6$

S_2 = sum of the minors of its leading diagonal elements

$$= \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$= (4 - 1) + (4 - 2) + (4 - 1) = 3 + 2 + 3 = 8$$

$$S_3 = |A| = 2(4 - 1) + 1(-2 + 1) + 2(1 - 2)$$

$$= 2(3) + 1(-1) + 2(-1) = 6 - 1 - 2 = 3$$

$\therefore$ The characteristic equation of A is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

$$(i.e.,) \quad \lambda^3 - 6\lambda^2 + 8\lambda - 3 = 0$$

By Cayley-Hamilton theorem

[Every square matrix satisfies its own characteristic equation]

(*i.e.,*) $A^3 - 6A^2 + 8A - 3I = O$ ... (1)

Verification :

$$A^2 = A \times A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix}$$

$$A^3 = A \times A^2 = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} = \begin{bmatrix} 29 & -28 & 38 \\ -22 & 23 & -28 \\ 22 & -22 & 29 \end{bmatrix}$$

$$\therefore A^3 - 6A^2 + 8A - 3I = \begin{bmatrix} 29 & -28 & 38 \\ -22 & 23 & -28 \\ 22 & -22 & 29 \end{bmatrix} - 6\begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix}$$

$$+ 8\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 29 & -28 & 38 \\ -22 & 23 & -28 \\ 22 & -22 & 29 \end{bmatrix} - \begin{bmatrix} 42 & -36 & 54 \\ -30 & 36 & -36 \\ 30 & -30 & 42 \end{bmatrix} + \begin{bmatrix} 16 & -8 & 16 \\ -8 & 16 & -8 \\ 8 & -8 & 16 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

To find A^4 :

$$(1) \Rightarrow \qquad A^3 = 6A^2 - 8A + 3I \qquad \ldots (2)$$

Multiply A on both sides, we get

$$A^4 = 6A^3 - 8A^2 + 3A = 6\,[6A^2 - 8A + 3\,I] - 8\,A^2 + 3A \qquad \text{by } (2)$$

$$= 36A^2 - 48A + 18\,I - 8A^2 + 3A$$

$$A^4 = 28A^2 - 45A + 18\,I \qquad \ldots (3)$$

$$(3) \Rightarrow A^4 = 28\begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} - 45\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 18\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 196 & -168 & 252 \\ -140 & 168 & -168 \\ 140 & -140 & 196 \end{bmatrix} - \begin{bmatrix} 90 & -45 & 90 \\ -45 & 90 & -45 \\ 45 & -45 & 90 \end{bmatrix} + \begin{bmatrix} 18 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 18 \end{bmatrix}$$

$$= \begin{bmatrix} 124 & -123 & 162 \\ -95 & 96 & -123 \\ 95 & -95 & 124 \end{bmatrix}$$

To find A^{-1} :

$$(1) \times A^{-1} \Rightarrow A^2 - 6A + 8I - 3A^{-1} = O$$

$$3A^{-1} = A^2 - 6A + 8I$$

$$3A^{-1} = \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} - 6\begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 8\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -6 & 9 \\ -5 & 6 & -6 \\ 5 & -5 & 7 \end{bmatrix} + \begin{bmatrix} -12 & 6 & -12 \\ 6 & -12 & 6 \\ -6 & 6 & -12 \end{bmatrix} + \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$$3A^{-1} = \begin{bmatrix} 3 & 0 & -3 \\ 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 0 & -3 \\ 1 & 2 & 0 \\ -1 & 1 & 3 \end{bmatrix}$$

Problem-5

Verify that $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ satisfies its own characteristic equation and hence find A^4.

Sol. Given : $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

The characteristic equation of the given matrix is $|A - \lambda I| = 0$

i.e., $\lambda^2 - S_1\lambda + S_2 = 0$

where S_1 = sum of the main diagonal elements

$= (1) + (-1) = 0$

$S_2 = |A| = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -1 - 4 = -5$

$\therefore$ The characteristic equation is $\lambda^2 - 0\lambda - 5 = 0$

i.e., $\lambda^2 - 5 = 0$

To prove : $A^2 - 5I = O$

$$A^2 = A \times A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4 & 2-2 \\ 2-2 & 4+1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$A^2 - 5I = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \qquad \dots (1)$$

$\therefore$ The given matrix A satisfies its own characteristic equation.

To find A^4 : From (1), we get $A^2 - 5I = O$

$A^2 = 5I$ multiply A^2 on both sides

$$A^4 = A^2(5I)$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix}$$

Problem-6

If $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$, then find A^n interms of A and I.

Sol. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0$

i.e., $\lambda^2 - S_1\lambda + S_2 = 0$ where

S_1 = Sum of the main diagonal elements

$= 1 + 2 = 3$

$$S_2 = |A| = \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} = 2 - 0 = 2$$

Therefore, the chracteristic equation of A is

$$\lambda^2 - 3\lambda + 2 = 0$$

i.e., $(\lambda - 2)(\lambda - 1) = 0$

i.e., $\lambda = 2, \lambda = 1$

Hence, the Eigenvalues of A are 1, 2.

To find A^n

When λ^n is divided by $\lambda^2 - 3\lambda + 2$

let the quotient be $Q(\lambda)$ and remainder be $a\lambda + b$.

$$\lambda^n = (\lambda^2 - 3\lambda + 2)\, Q(\lambda) + a\lambda + b \qquad \dots (1)$$

when $\lambda = 1$	when $\lambda = 2$
$1^n = a + b$	$2^n = 2a + b$

$$2a + b = 2^n \qquad \dots (2)$$

$$a + b = 1^n \qquad \dots (3)$$

Solving (2) and (3), we get

$(2) - (3) \Rightarrow a = 2^n - 1^n$

$(2) - 2 \times (3) \Rightarrow b = -2^n + 2(1^n)$

i.e., $a = 2^n - 1^n$

$b = 2(1^n) - 2^n$

Since, $A^2 - 3A + 2I = O$ by Cayley-Hamilton Theorem

$(1) \Rightarrow A^n = aA + bI$

$$A^n = (2^n - 1^n)A + [2(1^n) - 2^n]I$$

Problem-7

Use Cayley - Hamilton theorem to find the value of the matrix given by (i) $f(A) = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

(ii) $A^8 - 5A^7 + 7A^6 - 3A^5 + 8A^4 - 5A^3 + 8A^2 - 2A + I$ if the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

Sol. The characteristic equation of A is $|A - \lambda I| = 0$

i.e., $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$ where

S_1 = Sum of the main diagonal elements

$= 2 + 1 + 2 = 5$

S_2 = Sum of the minors of the main diagonal elements

$$= \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix}$$

$$= (2 - 0) + (4 - 1) + (2 - 0) = 2 + 3 + 2 = 7$$

$S_3 = |A|$

$$= \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$$

$$= 2\,(2 - 0) - 1\,(0 - 0) + 1\,(0 - 1) = 4 - 1 = 3$$

Therefore, the characteristic equation is

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By C - H theorem, we get

$$A^3 - 5A^2 + 7A - 3I = O \qquad \ldots (1)$$

Let $\quad f(A) = A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

$\quad g(A) = A^8 - 5A^7 + 7A^6 - 3A^5 + 8A^4 - 5A^3 + 8A^2 - 2A + I$

(i)

$$\begin{array}{r|l}
 & \lambda^5 + \lambda \\ \hline
\lambda^3 - 5\lambda^2 + 7\lambda - 3 & \lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 + \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda + 1 \\
 & \underline{\lambda^8 - 5\lambda^7 + 7\lambda^6 - 3\lambda^5 \qquad\qquad\qquad\qquad\qquad\qquad} \\
 & (-) \qquad\qquad\qquad \lambda^4 - 5\lambda^3 + 8\lambda^2 - 2\lambda \\
 & \qquad\qquad\qquad\quad \underline{\lambda^4 - 5\lambda^3 + 7\lambda^2 - 3\lambda} \\
 & \qquad\qquad (-) \qquad\qquad\qquad \lambda^2 + \lambda + 1
\end{array}$$

$$f(A) = (A^3 - 5A^2 + 7A - 3I)(A^5 + A) + A^2 + A + I$$

$$= O + A^2 + A + I \qquad \text{by (1)}$$

$$= A^2 + A + I \qquad \ldots (2)$$

Now, $$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$(2) \Rightarrow f(A) = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

(ii)

$$
\begin{array}{r|l}
 & \lambda^5 + 8\lambda + 35 \\
\hline
\lambda^3-5\lambda^2+7\lambda-3 & \lambda^8-5\lambda^7+7\lambda^6-3\lambda^5+8\lambda^4-5\lambda^3+8\lambda^2-2\lambda+1 \\
 & \lambda^8-5\lambda^7+7\lambda^6-3\lambda^5 \\
\hline
 & (-)\quad 8\lambda^4-5\lambda^3+8\lambda^2-2\lambda \\
 & 8\lambda^4-40\lambda^3+56\lambda^2-24\lambda \\
\hline
 & (-)\quad 35\lambda^3-48\lambda^2+22\lambda+1 \\
 & 35\lambda^3-175\lambda^2+245\lambda-105 \\
\hline
 & (-)\quad 127\lambda^2-223\lambda+106
\end{array}
$$

$$g(A) = (A^3 - 5A^2 + 7A - 3I)(A^5 + 8A + 35I) + 127A^2 - 223A + 106\,I$$

$$= O + 127A^2 - 223A + 106\,I$$

$$= 127\,A^2 - 223A + 106\,I$$

$$= 127A^2 - 223A + 106I$$

$$= 127\begin{bmatrix}5 & 4 & 4\\0 & 1 & 0\\4 & 4 & 5\end{bmatrix} - 223\begin{bmatrix}2 & 1 & 1\\0 & 1 & 0\\1 & 1 & 2\end{bmatrix} + 106\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$g(A) = \begin{bmatrix}295 & 285 & 285\\0 & 10 & 0\\285 & 285 & 295\end{bmatrix}$$

Problem-8

Use Cayley - Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ to express as a linear polynomial in A

(i) $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10\,I$

(ii) $A^4 - 4A^3 - 5A^2 + A + 2\,I$

Sol. Given : $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

The characteristic equation of A is $|A - \lambda I| = 0$

i.e., $\lambda^2 - S_1 \lambda + S_2 = 0$ where

S_1 = Sum of the main diagonal elements

$= 1 + 3 = 4$

$S_2 = |A| = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = 3 - 8 = -5$

$\therefore$ The characteristic equation of A is $\lambda^2 - 4\lambda - 5 = 0$

By Cayley - Hamilton theorem, we get

$$A^2 - 4A - 5I = O \qquad \dots (1)$$

To find : (i) $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10\,I$

$$
\begin{array}{r|l}
 & \lambda^3 - 2\lambda + 3 \\
\hline
\lambda^2 - 4\lambda - 5 & \lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10 \\
 & \underline{\lambda^5 - 4\lambda^4 - 5\lambda^3} \\
 & \qquad -2\lambda^3 + 11\lambda^2 - \lambda \\
 & \qquad \underline{-2\lambda^3 + 8\lambda^2 + 10\lambda} \\
 & \qquad\qquad 3\lambda^2 - 11\lambda - 10 \\
 & \qquad\qquad \underline{3\lambda^2 - 12\lambda - 15} \\
 & \qquad\qquad\qquad \underline{\lambda + 5}
\end{array}
$$

$$\therefore\ A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$$

$$= (A^2 - 4A - 5I)(A^3 - 2A + 3I) + A + 5I$$

$$= O + A + 5I = A + 5I \quad \text{by (1)}$$

which is a linear polynomial in A.

To find : (ii) $A^4 - 4A^3 - 5A^2 + A + 2I$

$$
\begin{array}{r|l}
 & \lambda^2 \\ \hline
\lambda^2 - 4\lambda - 5 & \lambda^4 - 4\lambda^3 - 5\lambda^2 + \lambda + 2 \\
 & \lambda^4 - 4\lambda^3 - 5\lambda^2 \\ \cline{2-2}
 & (-) \qquad\qquad\qquad \lambda + 2
\end{array}
$$

$$
\begin{aligned}
\therefore A^4 - 4A^3 - 5A^2 + A + 2I &= A^2(A^2 - 4A - 5I) + A + 2I \\
&= O + A + 2I \qquad \text{by (1)} \\
&= A + 2I
\end{aligned}
$$

which is a linear polynomial in A.

Home Work

1. Verify Cayley - Hamilton Theorem and find its inverse.

(a) $\begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$ Ans. $A^{-1} = \frac{1}{3}\begin{bmatrix} -3 & -2 & 2 \\ 6 & 5 & -2 \\ -6 & -2 & 5 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ Ans. $A^{-1} = \frac{1}{18}\begin{bmatrix} 7 & 1 & -1 \\ 1 & 4 & -2 \\ 1 & 7 & 7 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$ Ans. $A^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1/2 & -2 \\ 0 & 0 & 1/2 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ Ans. $A^{-1} = \frac{1}{4}\begin{bmatrix} 12 & 4 & 2 \\ -5 & -1 & -3 \\ -1 & -1 & -1 \end{bmatrix}$

(e) $\begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$ Ans. $A^{-1} = \frac{1}{11}\begin{bmatrix} 5 & -1 & -7 \\ -4 & 3 & 10 \\ 3 & -5 & -2 \end{bmatrix}$

2. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then prove that $A^3 - 3A^2 - 9A - 5I = O$

Hence, find A^4. **Ans.** $A^4 = \begin{bmatrix} 209 & 208 & 208 \\ 208 & 209 & 208 \\ 208 & 208 & 209 \end{bmatrix}$

3. Find A^n using Cayley - Hamilton theorem, taking

$A = \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}$ also find A^3.

Ans. $A^n = \left[\frac{9^n - 4^n}{5}\right] \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix} + \left[\frac{9.4^n - 4.9^n}{5}\right] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\therefore A^3 = \begin{bmatrix} 463 & 266 \\ 399 & 330 \end{bmatrix}$

4. Calculate A^4 for the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{bmatrix}$ **Ans.** $\begin{bmatrix} 16 & 32 & 577 \\ 180 & 16 & 609 \\ 0 & 0 & 625 \end{bmatrix}$

5. Verify Cayley - Hamilton theorem for the matrix

(i) $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

6. Using Cayley - Hamilton theorem, compute A^3 for $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

Ans. $\begin{bmatrix} 41 & 84 \\ 42 & 83 \end{bmatrix}$

7. Given that $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$, Express

$A^6 - 5A^5 + 8A^4 - 2A^3 - 9A^2 + 35A + 6I$ as a linear polynomial in A, using Cayley Hamilton theorem.

Ans. $4A + 42\,I$

3. ORTHOGONAL MATRICES

Definition:

A $n \times n$ square matrix $\boldsymbol{Q}$ is said to be an orthogonal matrix if its n column and row vectors are orthogonal unit vectors. **More specifically, when its column vectors have the length of one, and are pairwise orthogonal; likewise for the row vectors.**

The $n \times n$ square matrix Q is an orthogonal matrix if $Q \cdot Q^T = Q^T \cdot Q = I$, where I denotes the identity matrix

The inverse matrix of Q is Q^T

$$Q^{-1} = Q^T$$

Orthogonal Matrix Properties

- We can get the orthogonal matrix if the given matrix should be a square matrix.
- The orthogonal matrix has all real elements in it.
- All identity matrices are orthogonal matrices.
- The product of two orthogonal matrices is also an orthogonal matrix.
- The collection of the orthogonal matrix of order n x n, in a group, is called an orthogonal group and is denoted by 'O'.

- The transpose of the orthogonal matrix is also orthogonal. Thus, if matrix A is orthogonal, then is A^T is also an orthogonal matrix.
- In the same way, the inverse of the orthogonal matrix, which is A^{-1} is also an orthogonal matrix.
- The determinant of the orthogonal matrix has a value of ±1.
- It is symmetric in nature
- If the matrix is orthogonal, then its transpose and inverse are equal
- The eigenvalues of the orthogonal matrix also have a value of ±1, and its eigenvectors would also be orthogonal and real.

Problem-1

Verify that

$$A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \text{ is orthogonal.}$$

Solution:

$$A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix} \therefore\ A' = \frac{1}{3}\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$$

$$AA' = \frac{1}{9}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}\begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence,
A is an orthogonal matrix.

Problem-2

Determine if A is an orthogonal matrix.

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Solution: To find if A is orthogonal, multiply the matrix by its transpose to get identity matrix.

Given,

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Transpose of A,

$$A^T = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Now multiply A and A^T

$$\mathrm{A}\ A^T = \begin{bmatrix} (-1)(-1) & (0)(0) \\ (0)(0) & (1)(1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Since, we have got the identity matrix at the end, therefore the given matrix is orthogonal.

Problem-3

Prove Q = $\begin{bmatrix} cosZ & sinZ \\ -sinZ & cosZ \end{bmatrix}$ is orthogonal matrix.

Solution:

Given, Q = $\begin{bmatrix} cosZ & sinZ \\ -sinZ & cosZ \end{bmatrix}$

So, $Q^T = \begin{bmatrix} cosZ & -sinZ \\ sinZ & cosZ \end{bmatrix}$ ....(1)

Now, we have to prove $Q^T = Q^{-1}$

Now let us find Q^{-1}.

$$Q^{-1} = \frac{Adj(Q)}{|Q|}$$

$$Q^{-1} = \frac{\begin{bmatrix} cosZ & -sinZ \\ sinZ & cosZ \end{bmatrix}}{cos^2Z + sin^2Z}$$

$$Q^{-1} = \frac{\begin{bmatrix} cosZ & -sinZ \\ sinZ & cosZ \end{bmatrix}}{1}$$

$Q^{-1} = \begin{bmatrix} cosZ & -sinZ \\ sinZ & cosZ \end{bmatrix}$ ...(2)

Now, compare (1) and (2), we get $Q^T = Q^{-1}$

Therefore, Q is an orthogonal matrix

Problem-4

Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

Solution

$$\text{Let } A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \text{ Then, } A^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}.$$

So, we get

$$AA^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$AA^T = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \; .$$

Similarly, we get $A^T A = I_2$. Hence $AA^T = A^T A = I_2 \Rightarrow \; A$ is orthogonal.

Problem-5

If $A = \frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A'$. A' being the transpose of A.

Solution. We have,

$$A = \frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}, \quad A' = \frac{1}{9}\begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

$$AA' = \frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \cdot \frac{1}{9}\begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

$$= \frac{1}{81}\begin{bmatrix} 64+1+16 & -32+4+28 & -8-8+16 \\ -32+4+28 & 16+16+49 & 4-32+28 \\ -8-8+16 & 4-32+28 & 1+64+16 \end{bmatrix}$$

$$= \frac{1}{81}\begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ or } AA' = I$$

$$A' = A^{-1}$$

Home Work

Verify whether the matrix $A = \frac{1}{3}\begin{bmatrix} 2 & 2 & 1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$ is orthogonal.

Verify that $\frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$ is an orthogonal matrix.

Show that $\begin{bmatrix} \cos\phi & 0 & \sin\phi \\ \sin\theta\sin\phi & \cos\theta & -\sin\theta\cos\phi \\ -\cos\theta\sin\phi & \sin\theta & \cos\theta\cos\phi \end{bmatrix}$ is an orthogonal matrix.

Show that $A = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$ is an orthogonal matrix.

4. ELEMENTARY TRANSFORMATIONS

Any one of the following operations on a matrix is called an elementary transformation.

1. Interchanging any two rows (or columns). This transformation is indicated by R_{ij}, if the ith and jth rows are interchanged.
2. Multiplication of the elements of any row R_i (or column) by a non-zero scalar quantity k is denoted by $(k.R_i)$.
3. Addition of constant multiplication of the elements of any row R_j to the corresponding elements of any other row R_j is denoted by $(R_i + kR_j)$.

If a matrix B is obtained from a matrix A by one or more E-operations, then B is said to be equivalent to A. The symbol ~ is used for equivalence.

$$i.e., A \sim B.$$

Problem-1

Reduce the following matrix to upper triangular form

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix}$$

Solution. *Upper triangular matrix.* If in a square matrix, all the elements below the principal diagonal are zero, the matrix is called an upper triangular matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 3 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & -5 & -7 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & -2 \end{bmatrix} R_3 \to R_3 + 5R_2 \qquad \textbf{Ans.}$$

Problem-2

Transform $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix}$ *into a unit matrix.*

Solution:

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 10 \\ 3 & 8 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & -2 & 4 \\ 0 & -1 & -5 \end{bmatrix} \begin{matrix} \\ R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & -2 \\ 0 & -1 & -5 \end{bmatrix} R_2 \to -\frac{1}{2} R_2 \sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & -7 \end{bmatrix} \begin{matrix} R_1 \to R_1 - 3R_2 \\ \\ R_3 \to R_3 + R_2 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} R_3 \to -\frac{1}{7} R_3 \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 \to R_1 - 9R_3 \\ R_2 \to R_2 + 2R_3 \\ \\ \end{matrix}$$

Home Work

Reduce the following matrices into an identity matrix.

1. $$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

2. $$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 3 \\ 3 & 0 & 4 & 1 \end{bmatrix}$$

5. REDUCTION TO NORMAL FORM

NORMAL FORM (CANONICAL FORM)

By performing elementary transformation, any non-zero matrix A can be reduced to one of the following four forms, called the Normal form of A :

(*i*) I_r (*ii*) $[I_r \ \ 0]$ (*iii*) $\begin{bmatrix} I_r \\ 0 \end{bmatrix}$ (*iv*) $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$

Problem-1

Reduce to normal form of the following matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{bmatrix}$

Solution:

Step (i):

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ \mathbf{0} & \mathbf{-3} & \mathbf{-2} & \mathbf{-5} \\ \mathbf{0} & \mathbf{-6} & \mathbf{-4} & \mathbf{-22} \end{bmatrix}$$

$$R_2 \Rightarrow R_2 - \left(\frac{2}{1}\right) R_1 = R_2 - 2R_1$$

$\boldsymbol{R_2 \Rightarrow R_2 - 2R_1}$	$2-2(1)$ $\mathbf{= 0}$	$1-2(2)$ $\mathbf{= -3}$	$4-2(3)$ $\mathbf{= -2}$	$3-2(4)$ $\mathbf{= -5}$

$$R_3 \Rightarrow R_3 - \left(\frac{3}{1}\right) R_1 = R_3 - 3R_1$$

$\boldsymbol{R_3 \Rightarrow R_3 - 3R_1}$	$3-3(1)$ $\mathbf{= 0}$	$0-3(2)$ $\mathbf{= -6}$	$5-3(3)$ $\mathbf{= -4}$	$-10-3(4)$ $\mathbf{= -22}$

Step (ii):

$$\begin{bmatrix}1 & 2 & 3 & 4\\ 0 & -3 & -2 & -5\\ 0 & -6 & -4 & -22\end{bmatrix} \sim \begin{bmatrix}1 & 2 & 3 & 4\\ 0 & -3 & -2 & -5\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{-12}\end{bmatrix}$$

$$R_3 \Rightarrow R_3 - \left(\frac{-6}{-3}\right)R_2 = R_3 - 2R_2$$

$\boldsymbol{R_3 \Rightarrow R_3 - 2R_2}$	$0 - 2(0) = \mathbf{0}$	$-6 - 2(-3) = \mathbf{0}$	$-4 - 2(-2) = \mathbf{0}$	$-22 - 2(-5) = \mathbf{-12}$

$$\begin{bmatrix}1 & 2 & 3 & 4\\ 0 & -3 & -2 & -5\\ 0 & 0 & 0 & -12\end{bmatrix} \sim \begin{bmatrix}1 & 2 & \mathbf{4} & \mathbf{3}\\ 0 & -3 & \mathbf{-5} & \mathbf{-2}\\ 0 & 0 & \mathbf{-12} & \mathbf{0}\end{bmatrix}$$

$$C_3 \Leftrightarrow C_4$$

Step (iii):

$$\begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & -3 & -5 & -2 \\ 0 & 0 & -12 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 0 & \mathbf{-3} & \mathbf{-5} & \mathbf{-2} \\ 0 & \mathbf{0} & \mathbf{-12} & \mathbf{0} \end{bmatrix}$$

$C_2 \Rightarrow C_2 - \left(\frac{2}{1}\right) C_1 = C_2 - 2C_1$	$C_3 \Rightarrow C_3 - \left(\frac{4}{1}\right) C_1 = C_3 - 4C_1$	$C_4 \Rightarrow C_4 - \left(\frac{3}{1}\right) C_1 = C_4 - 3C_1$
$2 - 2(1) = 0$ $-3 - 2(0) = -3$ $0 - 2(0) = 0$	$4 - 4(1) = 0$ $-5 - 4(0) = -5$ $-12 - 4(0) = -12$	$3 - 3(1) = 0$ $-2 - 3(0) = -2$ $0 - 3(0) = 0$

Step (iv):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & -5 & -2 \\ 0 & 0 & -12 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \mathbf{0} & \mathbf{0} \\ 0 & -3 & \mathbf{0} & \mathbf{0} \\ 0 & 0 & \mathbf{-12} & \mathbf{0} \end{bmatrix}$$

$C_3 \Rightarrow C_3 - \left(\frac{-5}{-3}\right) C_2 = C_3 - \left(\frac{5}{3}\right) C_2$	$C_4 \Rightarrow C_4 - \left(\frac{-2}{-3}\right) C_2 = C_4 - \left(\frac{2}{3}\right) C_2$
$0 - \left(\frac{5}{3}\right)(0) = 0$ $-5 - \left(\frac{5}{3}\right)(-3) = 0$ $-12 - \left(\frac{5}{3}\right)(0) = -12$	$0 - \left(\frac{2}{3}\right)(0) = 0$ $-2 - \left(\frac{2}{3}\right)(-3) = 0$ $0 - \left(\frac{2}{3}\right)(0) = 0$

Step (v):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & -12 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = [I_3\ 0]$$

$$R_2 \Longrightarrow \frac{R_2}{-3}$$

$$R_3 \Longrightarrow \frac{R_3}{-12}$$

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = [I_3\ 0]$ is the Normal form of A.

Problem-2

Reduce to normal form of the following matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$

Solution:

Step(i):

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & -7 & 4 & -7 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$R_2 \Rightarrow R_2 - \left(\frac{4}{1}\right) R_1 = R_2 - 4R_1$$

$$R_3 \Rightarrow R_3 - \left(\frac{3}{1}\right) R_1 = R_3 - 3R_1$$

$$R_4 \Rightarrow R_4 - \left(\frac{1}{1}\right) R_1 = R_4 - R_1$$

Step(ii):

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & -7 & 4 & -7 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$R_3 \Rightarrow R_3 - \left(\frac{-7}{-7}\right) R_2 = R_3 - R_2$$

Step(iii):

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 1 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_4 \Rightarrow R_4 - \left(\frac{1}{-2}\right) R_3 = R_4 + \left(\frac{1}{2}\right) R_3$$

Step(iv):

$$\begin{bmatrix}1 & 2 & -1 & 3\\ 0 & -7 & 6 & -11\\ 0 & 0 & -2 & 4\\ 0 & 0 & 0 & 0\end{bmatrix} \sim \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -7 & 6 & -11\\ 0 & 0 & -2 & 4\\ 0 & 0 & 0 & 0\end{bmatrix}$$

$$C_2 \Rightarrow C_2 - \left(\frac{2}{1}\right)C_1 = C_2 - 2C_1$$

$$C_3 \Rightarrow C_3 - \left(\frac{-1}{1}\right)C_1 = C_3 + C_1$$

$$C_4 \Rightarrow C_4 - \left(\frac{3}{1}\right)C_1 = C_4 - 3C_1$$

Step(v):

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 6 & -11 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -7 & 0 & 0 \\ 0 & 0 & -2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_3 \Rightarrow C_3 - \left(\frac{6}{-7}\right) C_2 = C_3 + \left(\frac{6}{7}\right) C_2$$

$$C_4 \Rightarrow C_4 - \left(\frac{-11}{-7}\right) C_2 = C_4 - \left(\frac{11}{7}\right) C_2$$

Step(vi):

$$\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -7 & 0 & 0\\ 0 & 0 & -2 & 4\\ 0 & 0 & 0 & 0\end{bmatrix} \sim \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -7 & 0 & 0\\ 0 & 0 & -2 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}$$

$$C_4 \Rightarrow C_4 - \left(\frac{4}{-2}\right) C_3 = C_4 + 2C_3$$